

PQC-Readiness Audit Report

Contract: /mnt/d/Crypto/audit_case_studies/contracts/OpenZeppelin_ECDSA.sol
SHA-256: 80da22a9603dc4410d8e3175527eb4e5... **Lines:** 113 **Date:** 2026-08-20 **Tool:** pqc-audit v0.2.0

Executive Summary

Check	Result	Detail
Security Scan	PASS	0 critical, 0 high
PQC Readiness	FAIL	2 quantum-vulnerable primitives, 0 critical implementation advisories
Gas Optimization	5 opportunities	~1000 gas units
OVERALL	FAIL	

Phase 1: Structure

- **Solidity Version:** ^0.8.0
- **Contracts:** 0
- **Functions:** 11
- **Events:** 0

Phase 2: Gas Optimization (5 found)

#	Pattern	Impact	Fixable
1	Use uint256 instead of smaller uints	Info	Manual
2	Use uint256 instead of smaller uints	Info	Manual
3	Use uint256 instead of smaller uints	Info	Manual
4	Use uint256 instead of smaller uints	Info	Manual
5	Use uint256 instead of smaller uints	Info	Manual

Phase 3: Security Analysis (1 findings)

Severity	Count
Critical	0
High	0
Medium	1
Low	0

Security Score: 95.0/100

Top Findings

- **[MEDIUM]** Line 75 — Missing zero address check
 - Code: `if (signer == address(0)) {`
 - Missing zero address check for critical operations.

Phase 4: Post-Quantum Cryptography Readiness

Line	Primitive	Risk	Migration Path
74	<code>ecrecover</code>	CRITICAL	Replace with ML-DSA-65 (Dilithium-3) via a FIPS 204 implemen. . .
92	<code>keccak256</code>	MEDIUM	Double hash output length or use SHA-3-512. . .
9	<code>ecdsa</code>	CRITICAL	ML-DSA-65 (FIPS 204). . .

Quantum Risk Level: CRITICAL

NIST Status: FIPS 204 (ML-DSA) and FIPS 205 (SLH-DSA) finalized. FIPS 206 (FN-DSA/Falcon) selected for standardization; publication pending.

Implementation Vulnerability Scan (2024-2026 Research)

Vulnerability	Risk	Source
<i>None detected</i>	LOW	No known implementation vulnerabilities match this contract

Evidence Attestation

- **Source SHA-256:** 80da22a9603dc4410d8e3175527eb4e520e4c24e6053bd3f2e04d1e5f4c0f201
- **Report SHA-256:** 568d28184189d199b5364a2ef3d4e331...

This report was generated mechanically. No LLM inference was used in the security or PQC assessments. PQC-Readiness Audit — pqc-audit v0.2.0